LYRA HUANG

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EDUCATION

University of Virginia
Expected May 2027

Charlottesville, Virginia | B.A. in Astronomy and B.A. in Computer Science; Minor in Philosophy

GPA: 3.754 / 4.00 (through Spring 2026)

RESEARCH EXPERIENCE

Undergraduate Researcher, Astronomy Senior Thesis
2026–Present

University of Virginia | Advisor: Prof. Phil Arras

- Investigating charged dust dynamics in exoplanetary magnetospheres using numerical models of stellar gravity, radiation pressure, Poynting–Robertson drag, and Lorentz forces.
- Developed Python and REBOUND-based simulations to examine particle trajectories, radial migration, and bounded versus escaping motion across magnetic-field strengths, grain sizes, and charge conditions.
- Implemented and compared radiation-pressure models with grain-size-dependent efficiency Q_pr and beta; analyzed sensitivity of outcomes to magnetic-field polarity and initial conditions.
- Evaluated numerical behavior with convergence checks, analytical cyclotron-frequency comparisons, parameter sweeps, and trajectory/time-series visualizations.

Undergraduate Research Tutorial | DREAMS Cosmological Simulations
Spring 2026

University of Virginia | ASTR 4993 | Advisor: Prof. Paul Torrey

- Investigated simulated galaxy properties using DREAMS cosmological simulation data, studying connections between simulation parameters and galaxy observables.
- Adapted Python research code and prepared simulation data for conditional normalizing-flow models; ran model-training experiments on UVA's Rivanna high-performance computing cluster.
- Compared training hyperparameters and evaluated generated versus simulated property distributions using plots and statistical checks; documented findings in a March 2026 research report.

PROFESSIONAL EXPERIENCE

Software Intern, Software Department
Summer 2026

VIA-tech | Shenzhen, China

- Model data: Contributed to a unified annotation-data pipeline connecting YOLO, COCO, Pascal VOC, and Label Studio formats; used round-trip tests to check conversion consistency.
- Camera calibration: Helped develop AprilTag-assisted point selection, camera-intrinsics import, bird's-eye-view coordinate editing, and MV3DT export workflows.
- Multi-camera tracking: Contributed to Python-based tracking infrastructure using MQTT real-time streams and timestamp synchronization across cameras.

SELECTED PROJECTS

Emberline | 2D Puzzle-Platformer Demo

Completed demo

Independent project | Godot 4 (GDScript) | Playable macOS and Windows builds

- Created a playable puzzle-platformer centered on carrying, throwing, and transferring a flame to activate light-sensitive and darkness-dependent mechanisms, bridges, and climbable vines.
- Implemented interactive level systems, player movement and climbing, checkpoints, hazards, audiovisual feedback, and a multi-stage level with a victory condition; included functional test scenes.

BeLie | 2D Puzzle Game Concept (Godot)

Fall 2026–Present

CS 4730, Computer Game Design | Team project, early development

- Co-developing a puzzle game concept in which players collect words through observation and combine them into belief statements that alter the world; one belief can be active at a time.
- Responsible for character design and map/terrain design during concept development, with Godot as the project engine.

MediCamp | Healthcare Visit Support App Prototype

HCI | Team of five | Project Website | Figma Prototype

- Managed milestones and team deliverables for a healthcare-support app prototype involving triage, appointment booking, navigation, and accessibility.
- Developed personas, task flows, wireframes, and a high-fidelity interactive Figma prototype; refined information architecture and interactions using critique and usability feedback.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, C, C++, SQL

Research and development: REBOUND, Rivanna HPC, Git/GitHub, Linux, Godot 4, GDScript, Figma, JUnit 5

Methods and tools: Numerical simulation, conditional normalizing flows, data visualization, camera calibration, computer vision annotation formats, MQTT, SQLite, Adobe Photoshop

TEACHING, OUTREACH & SERVICE

McCormick Observatory Public Nights Volunteer

Oct 2023–Present

University of Virginia

- Volunteer at public observing nights (approximately 0–2 events per semester, depending on scheduling); assist visitor flow and tours and explain astronomy concepts at events hosting around 120 attendees.

Astro Fun Club

Sep 2023–Present

University of Virginia

- Participate in astronomy observations, field activities, and science outreach.

HONORS & SELECTED COURSEWORK

Honors: Dean's List, University of Virginia (five semesters: Fall 2023, Spring 2024, Spring 2025, Fall 2025, Spring 2026).

Completed coursework: Human-Computer Interaction in Software Development; Astrophysics Research Methods; Observational Astronomy; Applied Linear Algebra; Data Structures and Algorithms; Introductory Cybersecurity; Discrete Mathematics.

Current coursework (Fall 2026): Artificial Intelligence; Machine Learning; Computer Game Design; Astronomy Senior Thesis; Minds, Machines, and Persons.